AN ETALE MORPHISM IS LOCALLY QUASI-FINITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-morphisms/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot. Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Editorial note. Shortest proof in the batch: a two-link chain of implications, each link named with its lemma. This is a complete (if trivial) proof by transitivity, deliberately distinguished from the class refused in this window – those leave a statement item with no argument at all ('we omit the proof of (3) and (4)'), while this span exhibits the two implications that compose to the assertion. It is also not the class of the sibling lane's skipped lemma-etale-smooth-unramified (source lines 8150-8159), whose entire proof is 'This follows immediately from the definitions.'

Allowed standard prerequisites (editorial).

- An etale morphism is unramified. Lemma \ref{lemma-etale-smooth-unramified}, cited at proof lines 8168-8169. First link of the chain.
- An unramified morphism is locally quasi-finite. Lemma \ref{lemma-unramified-quasi-finite}, cited at proof lines 8170-8171. Second link of the chain.

Source reference. Original lemma 03WS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5123–5123. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 02GK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3231–3231. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02V5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3756–3756. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *An étale morphism is locally quasi-finite.*

Proof. By Lemma 02GK an étale morphism is unramified. By Lemma 02V5 an unramified morphism is locally quasi-finite. □